Deep Learning Models for Gait Event Prediction

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***Abstract*—Research in human gait has contributed to clinical applications in physiotherapy and preventative medicine by improving the understanding and measurement of human gait patterns. In this context, characterization of gait events and kinematic models are essential for medical intervention. This paper proposes a novel gait event detection method, using deep learning for the kinematic model. The setup uses two IMU sensors from which the angles of the kinematic model are computed, and a FSR sensor to detect the Heel Strike event (HS). A microcontroller digitally processed the signals from these sensors in real-time, and saved the gait data. We collected data from healthy subjects representing various age groups (young, adolescent, and elderly), selected by a medical specialist. Firstly, a kinematic curve pattern was created using three types of deep networks. Secondly, the pattern was subsequently applied to detect Heel Strike (HS) events using Dynamic Time Warping (DTW). A detection average of 96.64% was obtained using a single kinematic model, compared to 76.82% obtained in other studies for HS detection. Furthermore, our research provides medical specialists with a kinematic gait curve to develop tools to automatically detect critical points in gait impairments.**

***Index Terms*—Healthcare system, Gait Events, IMU Sensors, Deep Learning, Dynamic Time Warping.**

HUMAN gait patterns are the emergent result of a concerted effort by the bones, tendons and muscles of the human body, directed by the brain. Gait analysis [1] involves the kinematic modeling and characterization of normal and anormal patterns in human gait, and is used by physiotherapists, orthopedists, and neurologists to assess the health status of patients and select medical treatments. While gait analysis can be challenging to use in an effective way [2], it is also a very important diagnostic and assessing tool in neurology, rehabilitation, and physical therapy to detect disorders and monitor neurological conditions [3]. Neurological conditions like the aftereffects of a stroke, accident, and Parkinson's disease can result in the deterioration of motor skills and a decrease in the ability to walk, significantly impacting the overall well-being and health of individuals [4]. In this context, there is a need to assist elderly individuals and patients to overcome movement disorders and restoring their ability to walk, ultimately enhancing their daily lives, and gait analysis can play a key role in this regard [5].

The human gait cycle can be analysed according to many parameters such as the range and pattern of joint motion, floor reaction forces, timing of foot/floor contact, gait speeds, cadences, stride lengths, swing-to-stance phase relations, resting joint angles, isometric torques, and phasic muscular activities [6]. From these, the swing-to-stance phase parameters feature three crucial events in each gait cycle: the Mid-Swing (MS), Heel-Strike (HS), and Toe-Off (TO) events. The events in the gait cycle can be recorded using non-wearable sensors, wearable sensors, and hybrid systems [7]. Of these, wearable sensors are most commonly used, since they are less expensive and can be used outside controlled environments. Typically used wearable sensors include accelerometers, gyroscopes, active markers, and Inertial Measurement Units (IMUs).

Many different mathematical techniques can be applied to analyse gait patterns from sensor signals. For instance, Staab *et al.* [8] used spectral analysis to detect osteoarthritis, while Liu *et al.* [9] developed an intelligent shoe with multiple sensors, using the continuous wavelet transform to identify gait events. Machine learning techniques can be also used for this purpose [10], and also combined with frequency analysis in order to reduce the computational load [11]. Machine learning can also be used to detect redundancies in data in order to minimize training time and the number of required sensors for gait analysis [12]. More recent deep learning techniques can also be used, such as training a LSTM network to detect HS events [13] or detect Parkison disease from spatiotemporal gait data [14], [15].

In this work, a Data Acquisition System (DAS) based on a Force-Sensing Resistor (FSR) and two IMUs is used to acquire a kinematic gait model, and a novel method that uses deep learning is proposed to detect gait events using a kinematic gait model, reducing the number of required sensors and providing feedback to users about gait quality. Data from the FSR sensor is used to train the kinematic model, and the

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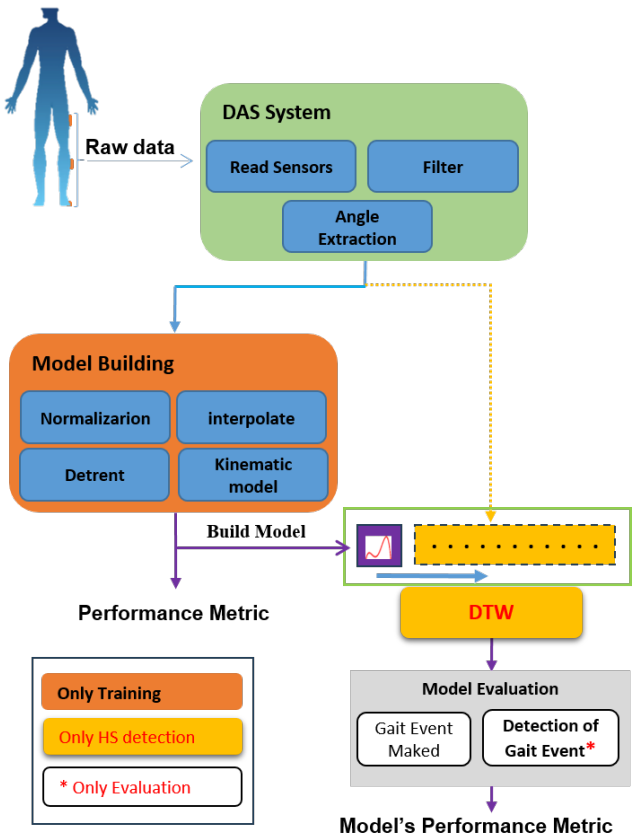

Fig. 1: Flow architecture of the proposed model

IMUs are then used to detect HS events using Dynamic Time Warping (DTW). To validate the generality of our method, we apply it to widely different gait data from teenagers, young adults and elder people. Additionaly, we compare our results in detection of HS events with Wang et al. [15].

I. Methodology

This section presents the method to detect HS gait events we propose. In each subject, two IMU sensors were installed on the shin and thigh, and an FSR was installed on the heel. From the signals collected from these sensors, the DAS performs online mathematical operations and extracts the knee angle through a recursive filter. The collected angles are grouped in a matrix to build a kinematic model of the knee using deep learning techniques. The model is stored to perform a DTW with the newly acquired data to classify the HS event. Finally, we evaluate our proposal with another methodology based on recursive neural networks. Fig. 1 illustrates the proposed workflow of the method.

A. Data Acquisition System (DAS)

In this work, a total of three sensors were installed on different parts (feet, shins, and thighs) of the legs. The IMUs on the shin and thigh contain a 3-axis accelerometer X, Y, and Z and a 3-axis gyroscope (a total of 12 signals). Moreover, the signals were collected at a sampling rate of 25Hz (40ms).

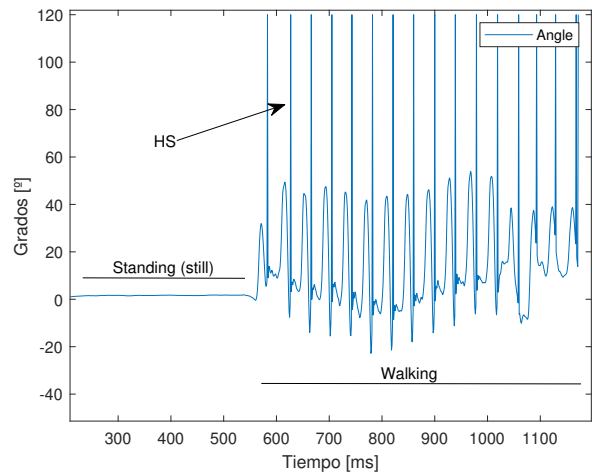

Fig. 2: Angle signal acquired from the DAS. In this example, the patient is first still and then starts walking.

An Atmega328P microcontroller digitally processes online the raw data from the sensors to extract the knee angle. The accelerometer signal is passed through a Kalman filter and converted to an angle using Eq. 1 [16]. For instance, to an angle in the X-axis:

$$A_{angleX} = \arctan\left(\sqrt{\frac{f_X^2}{f_Y^2 + f_Z^2}}\right) \tag{1}$$

where f_X, f_Y, f_Z are the gravity forces, i.e. the accelerometer signals. f_X is in the numerator since the principal signal would be the X-axis.

The gyroscope signal is integrated to obtain the angle. Discretization can take the integral into a differencial equation. Hence, integration can be approximated using trapezoidal integration [17]:

$$G_{angleX} = \int v_x(t)dt = \sum_{n=0}^{N}\left[\frac{V_x(n) + V_x(n-1)}{2}\right]T_s \tag{2}$$

where v_x, V_x are the angular velocity and T_s is sample time.

Finally, a complementary filter joins the accelerometer and gyroscope angles, and the FSR sends the HS event. To avoid erroneous heel contacts, an average of the FIR filter with a prior input of N is used. An example of the acquired signal is presented in Fig. 2.

B. Model Building

There are methods for model regression based on time, frequency, and time-frequency domain, which are usually applied. This work proposes a set of signal processing steps to get a kinematic model:

1) Detrending time series data: It is necessary to clarify that the walks were not performed on electric treadmills, thus varying the walking time. Also, healthy patients walking on slopes were taken intentionally causing a trend in order to assess the adaptability of our method.

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The time series is assumed to be an additive combination of a trend and other components [18].

$$X(t) = T(t) + S(t) \tag{3}$$

where $X(t)$ is the original time series, $T(t)$ is the trend, and $S(t)$ is the residue of the season component that also includes noise.

To detrend a time series, we must first estimate and subtract the trend component from the original data, leaving only the seasonal component. Moreover, to select the best detrending model (a constant model, a linear model, a Statsmodels model, and a moving average model were tested), we visually inspected the data in each applied model. We looked for any remaining trends, oscillations, or patterns affecting our data, and the Statsmodel was selected. Figure 3a shows a part of the signal acquired in the time range (250-1250), and the signal without trend can be observed.

2) Data Preprocesing: After applying the detrend, the time series is normalized in the range [0,100] according to the following formula [19]:

$$T_i = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{4}$$

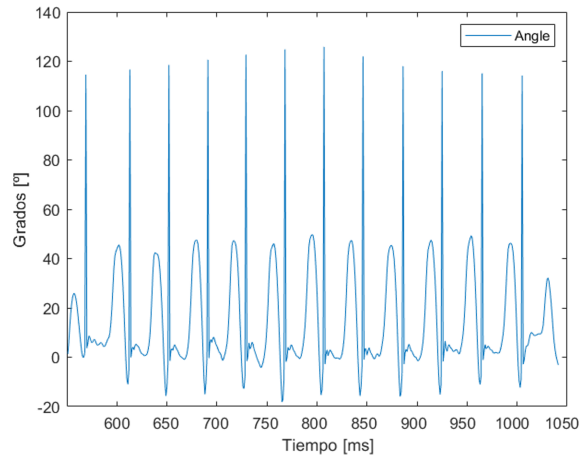
where T_i and x_i are the time before and after normalizing, and x_{min} and x_{max} are the minimum and maximum values of time selected.

3) Model building: To ensure that the model adapts to the variation in acquired data, including parameters that modify gait, such as age, gender, personal traits, dominant leg, and external factors, we aim to implement a model using deep learning techniques. Three deep learning models were created and experimented with different hyperparameters to know the performance.

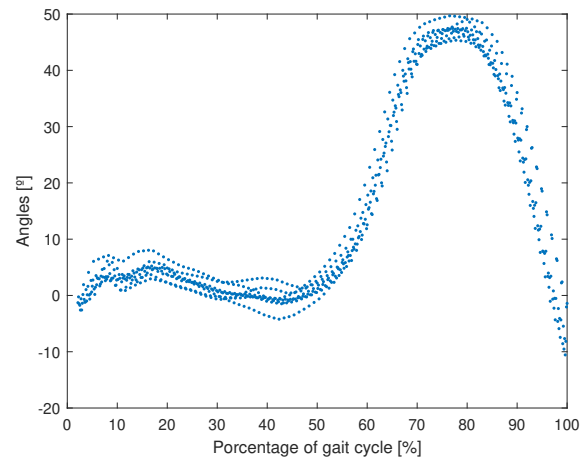
The first neural network model used was a Deep Neural Network (DNN). The model has four hidden layers: the first with 200 units and a sigmoid activation function; the second, third, and fourth with 100 units and a Relu activation function; and finally, the output layer has a linear activation function. We applied mean square error as a loss function and adaptive moment estimation as an optimization function. Finally, we trained it for 500 epochs. The results can be seen in the Fig. 4a.

Another model used was a Recursive Neural Network (RNN) in which we provided input data $X \in \mathbb{R}^{800\times1}$. The model comprises 32 RNN layers with a sigmoid activation function and 16 hidden layers with a ReLU activation function. Mean square error was applied as the loss function, and root mean square propagation was used as the optimization function. Finally, we trained it for 800 epochs. The results can be observed in the Fig. 4b.

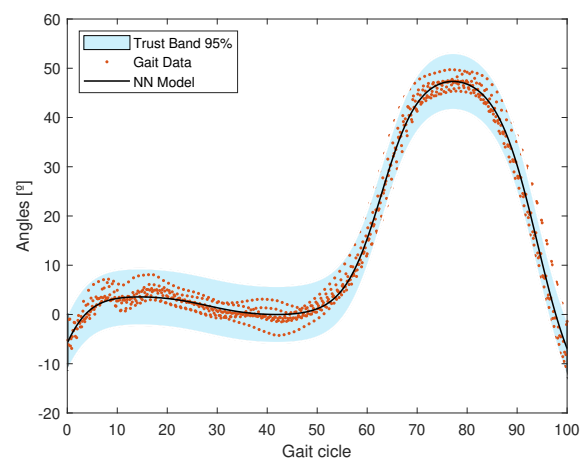
Finally, LSTM is used; the model comprises 32 LSTM layers with a sigmoid activation function. The model has three hide layers: the first and second with eight layers and a Relu activation function. Mean square error was applied as the loss function, and root mean square propagation was used as the optimization function. Finally, we trained it for 1000 epochs. The results are in the Fig. 4c. The resulting model is presented in detail in Fig. 3.



(a) Signal detrending



(b) Signal normalization and interpolation



(c) DNN Kinematic model and true Band

Fig. 3: Processing of the Human Gait Signal: a) Detrending signal of all the gaits performed by the person, b) Collection, normalization and interpolation of all the gaits in a set, and c) Modeling of the Kinematics of the Human Gait with a 95% True Band

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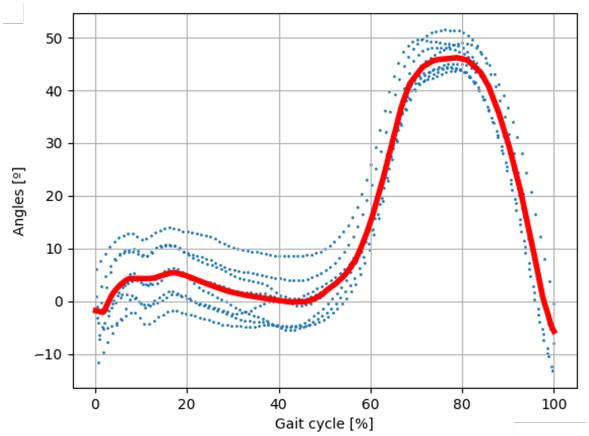
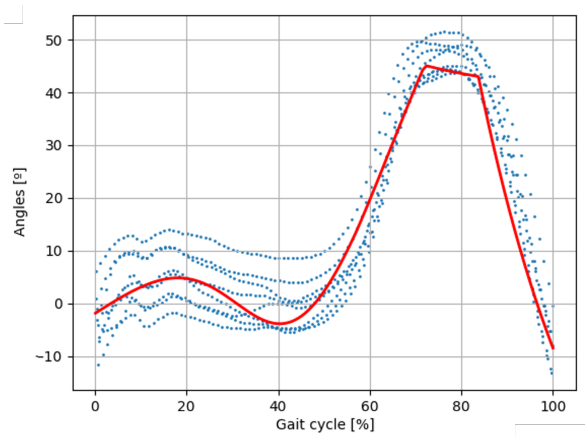
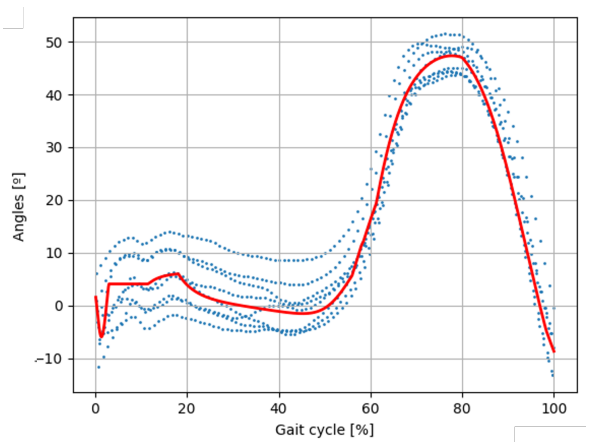


(a)



(b)



(c)

Fig. 4: Modeling Data Acquired with High Variance from the DAS to Represent Human Gait Kinematics Using Deep Learning Techniques: a) Deep Neural Network - DNN, b) Recursive Neural Network - RNN, and c) LSTM.

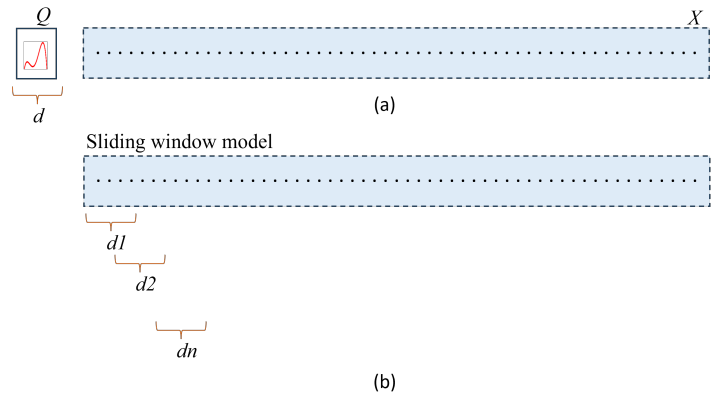


Fig. 5: Slide windows between query vector Q and the data collected X. a) Vector data representation, and b) sliding windows model.

4) Identifying similar signals: Assuming the gait data collected is saved in a vector X, and the selected kinematic model is saved in a vector Q, the latter will be a query to search in X. Therefore, the query Q is slid over X; that is, Q is shifted by one sample to the right each time sample while calculating the error measures in each interval.

In Fig. 5a the vectors Q and X are shown, and additionally, the size of the window (d) to be slid is represented. Fig. 5b indicates the calculation of the error measures d_i between Q and each window of the slid vector X, and each result is stored in a vector D.

$$D = [d_1, d_2, d_3, ..., d_{n-m+1}] \tag{5}$$

where n is X length, and m is the length of the query.

Dynamic Time Warping (DTW) algorithm allows us to stretch and alignment the similarity between two data sets that may vary over time, such as the subsequence of X data chosen for slide windows and the kinematic model Query. DTW has the advantage of comparing two signals even if there is a slight shift in time, in contrast to other methods of calculating the distance between signals. To measure the distance between two data the Euclidean Distance is used.

To identify the HS event, Algorithm 1 is used. In our work, z-normalization is used for both data, detailed in the following equation for X.

$$\hat{x} = \frac{x_i - \mu_X}{\sigma_X} \tag{6}$$

where μ is the mean and, σ is standard deviation.

By including Eq. 6, the error can be expressed in a single equation that represents the z-Normalized Euclidean distance:

$$d_{mn}(X, Y) = \sqrt{\sum_{k=1}^{k} (\hat{x}_k - \hat{y}_k)^2} \tag{7}$$

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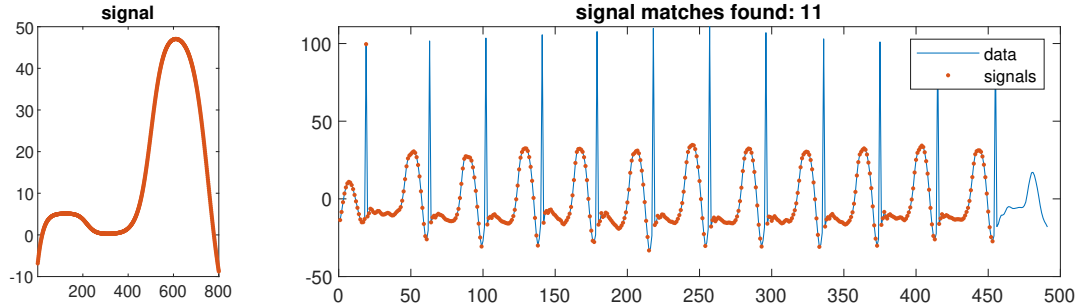


Fig. 6: Signal search with slide windows between the query vector and the collected data.

Algorithm 1 HS event detect algorithm, Q = query

Input: X_t = angle signal from the DAS,
Output: HS detect signals
Method: HS_event

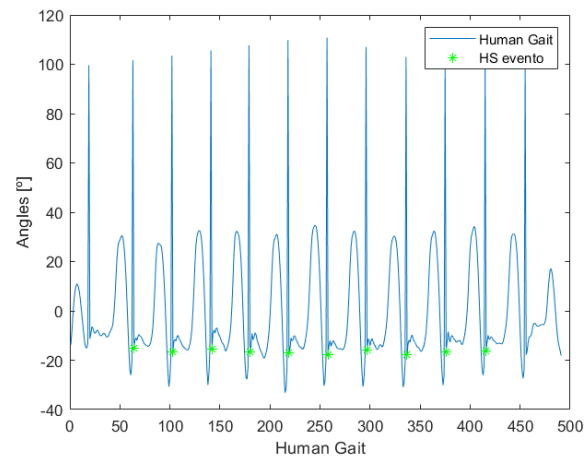
```
locate ← 1
while slide=1,2,...,length(X) do
    X_iter ← DTW(X_subsequence, Q)
    d_slide ← distance(X_iter, Q)
    if d_slide is correct distance then
        HS_locate ← Locate and save HS event point
        locate ← locate + 1
    end if
end while
return: HS
```

where d_{mn} drescribes the distance between the m-th samples of X and the n-th samples of Y, also, X and Y are k-dimensional signals.

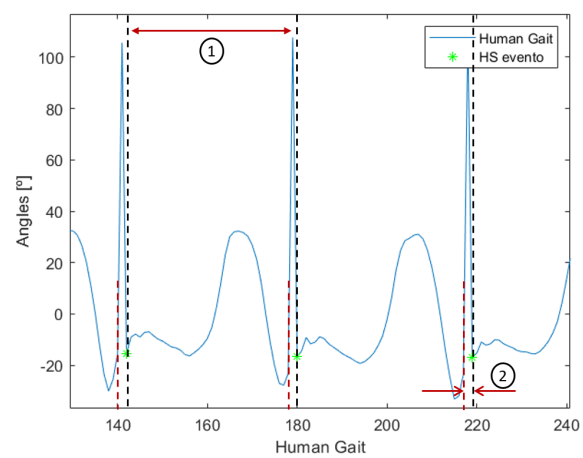
Fig. 6 presents the data traces and highlights any identified signal instances while the query slid over the signal. Since they are real vectors, the function shows the relationship based on the samples. Notice that the last gait in the range [450, 500] was not identified because it is outside the confidence band, suggesting a difference in the signal. Furthermore, it is noted that the first gait was identified as a signal.

II. Results

To assess the efficacy of our methodology, we propose a comparison with a study conducted by Wang et al. from the Department of Mechanical Engineering at the National Taiwan University [15]. In his research, Wang et al. developed a model using a LSTM network for detecting HS events in human gait with two IMUs placed on the leg, each one capturing three axial accelerations and three axial angular velocities. During the training phase, data acquired from the IMUs was manually labelled with the HS events. Data segmentation was used for all trainings, with sliding windows of 50 samples from the sensor data. The model input feed to the LSTM was represented as $X_{in} \in \mathbf{R}^{50x6}$. The output of the LSTM was fed to a fully-connected layer with a sigmoid activation to generate an output $P \in \mathbf{R}^{50x1}$, so that each output value was intended to be 1 at HS events and 0 otherwise. Wang et al.



(a) HS event detected in gait set



(b) Enlarged representation of Hs event detected in gait set that includes (1) the complete gait cycle and, (2) the time error in event detection

Fig. 7: Automatic detection of HS events a) Detection in all data sets and b) Zoom view to appreciate the detection of the event.

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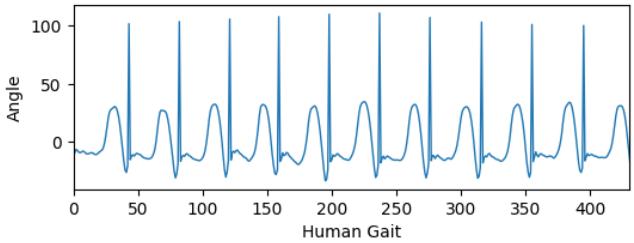
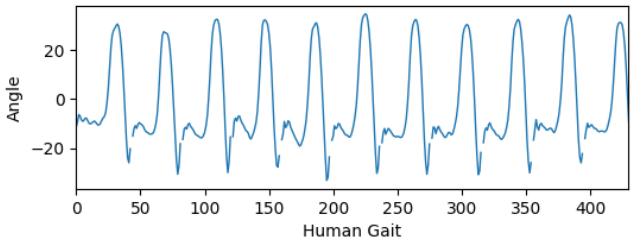
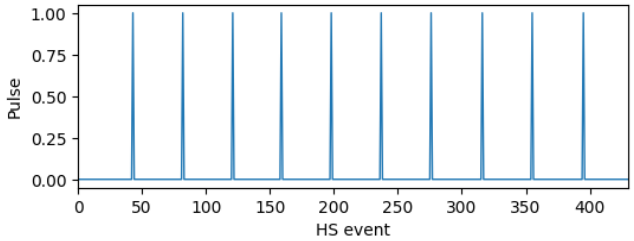


(a) Signal acquired from DAS, with FSR pulses



(b) Data without FSR pulses



(c) HS events manually generated from the FSR pulses.

(d) HS event prediction

Fig. 8: Data pipeline to train a LSTM model to predict HS events, using our methodology. First, a signal is acquired from the DAS (a), including HS events as detected by the FSR, visible as spikes. The spikes are removed from the original signal (b), and manually processed into 0-1 pulses, signaling HS events (c). The LSTM learns to predict HS events (d).

applied binary cross-entropy as a loss function to train their model with the manually labelled HS events, using Rmsprop as the optimizer. All trainings were run for 60 epochs with a batch size of 500, and a probability of 0.2 in the dropout layers; for validation we used Leave-One-Out Cross-Validation (LOOCV).

A. Dataset

Ten subjects selected by a medical specialist underwent a physical examination that verified that they had a normal

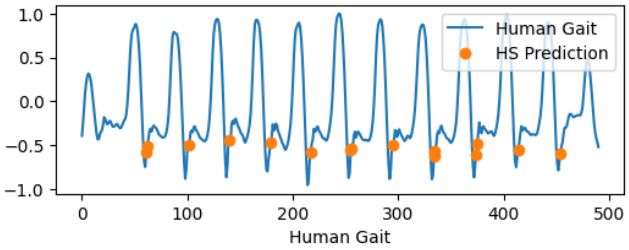


Fig. 9: Sometimes, the LSTM model predicts two HS events in rapid succession. Close neighbours are filtered out to remove this kind of error.

TABLE I: The characteristics of the subjects. Four male subjects and Six female subject participated in the experiment

Subject ID	Sex	Age	Height	Weigth
1	F	50	1.6	67.3
2	M	49	1.74	76.2
3	M	80	1.69	71.4
4	M	68	1.71	72.4
5	F	45	1.58	65.9
6	F	22	1.64	70
7	M	36	1.6	60
8	M	70	1.7	65.3
9	M	49	1.71	69.3
10	F	38	1.68	68.6

gait without any alteration, a positive attitude, and no visible or internal lesions in the lower limbs, with joints in good condition and without any alteration or affection. Each patient was tested to ensure that their gait was not unusual in any way in all three phases of gait.

The measured gait speed in the subjects was within a range of 0.9 to 1.4 meters per second. In older individuals, the decrease in step length had a significant impact on reducing speed than changes in cadence (in agreement with the clinical literature Pirker [20]), as well as an increase in time spent in the double support phase. These gait changes can be attributed to the normal aging process. However, it is essential to highlight that gait speed in older adults is a relevant indicator of their overall health and survival.

Table I presents the characteristics of the selected subjects. According to the World Health Organization (WHO), subjects are classified into the following categories: ”children” for individuals aged 0 to 17 years, ”young adults” for those between 18 and 65 years old, ”middle-aged” for individuals aged 66 to 79 years, ”elderly” for those between 80 and 99 years old, and ”oldest-old” for individuals over 100 years old [21].

B. Data Preparation

In the original setup in Wang et al., all data from the IMUs was used, but our DAS provides only one data point per time step, corresponding to the knee angle. Consequently, proper preparation of our data (Figure 8a) is essential. First, the data from the FSR sensor is removed (Figure 8b). Then, HS events are manually marked at each time step where a HS event can be inferred from the spike from the FSR sensor (Figure 8c);

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this is the ground truth for the output of the model, used to training and validation.

An example applying Wang's method to our data is presented in Figure 8d. Sometimes, the model erroneously predicts two HS events in rapid succession (Figure 9), leading to additional false positives. This problem was addressed by removing the nearest neighbours of the main pulse. This resulted in an improvement in the predictions, achieving an average detection delay of 48 ms.

C. Performance Evaluation

Following Wang et al. [15], we measure the performance of the model using the Positive Predictive Value (PPV):

$$PPV(\%) = \frac{n}{N} \times 100(\%) \tag{8}$$

Where N is the number of total steps in the input and n represents the number of HS events correctly detected by the model.

In a first experiment, gait data was separated in two subsets, one for males and another for females. For each subset two models were trained, one using our methodology, and the other using the methodology from Wang et al. In this way, we can compare the performance of both models to predict HS events. Results are shown in Figures 10a and 10b for the women and men subsets, respectively. In both figures, PPV-W and PPV-Our represent performance results from the model of Wang et al. and ours, respectively. Also, in both figures, performance values are grouped by subject, gender and left/right leg. For example, the label 6FL in Figure 10a marks the performance values of both models for the left leg of the sixth subject (from Table I), and 6FR for the right leg of the same subject.

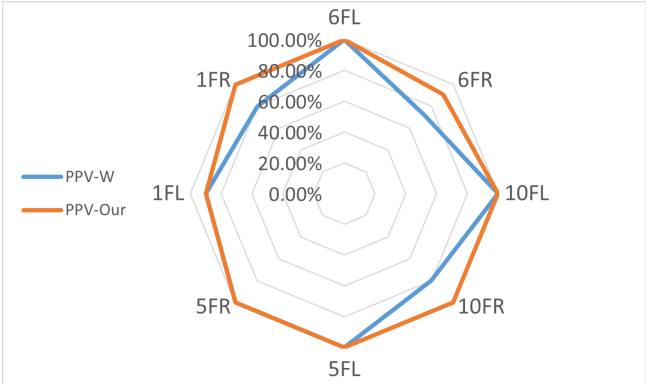
In both Figure 10a and 10b, results from the model of Wang et al. are very good, detecting all HS events from the left leg of all subjects, but lower detection rates are observed for the right leg in many cases; this aligns with the suggestion to use a different model for each leg from Wang et al. Quantitatively, the model of Wang et al. achieves a detection rate of HS events of 90.34% in women and 76.44% for men, while our method achieves 97.61% and 94.39%, respectively.

In a second experiment, all gait data from both women and men was grouped together in a single dataset to train our model and the model of Wang et al, and the results (Figure 11) were again tested separately for women and men. For men, both models generate almost the same results as in the previous experiment (training different models for women and men), with Wang et al. achieving good results for left legs but not right legs (detecting 76.44% of the HS events), and our model working for all legs (detecting 95.23% of the HS events). For women, however, our model detects 98.75% of all HS events, compared to 77.39% for Wang et al., so our model significantly outperforms Wang et al. for this case of this experiment.

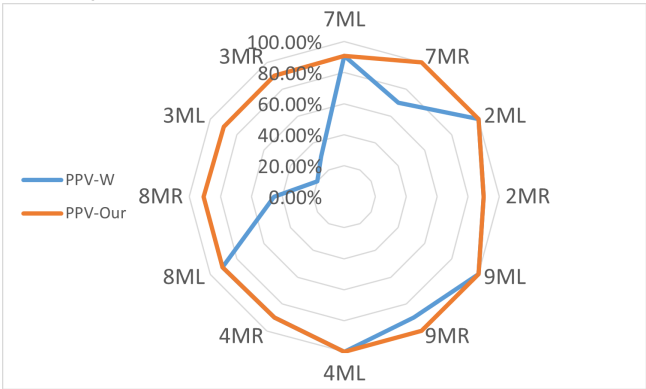
III. Conclusion

This work presented the detection of the heel strike event through a kinematic model developed using deep learning algorithms. Identifying this event translates into reducing sensor dependency, as in our case, eliminating the need to use FSR sensors. Acquiring a patient's gait model can be crucial for specialists as it facilitates quantification through comparison with a standard kinematic model, evaluating impairments in phases of gait, and measuring progress during a rehabilitation process.

Deep Learning methods such as DNN, RNN, and LSTM were applied. It is essential to highlight that the experimental data were acquired on flat terrain without obstacles, aiming to obtain a more representative model of everyday situations. The DNN model achieved the ability to detect events in both men's and women's gait, covering an age range from 18 to 80 years old and using a single model. We obtained an average detection rate of 96.64%, with an average detection latency of approximately 20 ms. In contrast, when comparing our method with a similar work using a similar data acquisition system [7],



(a) Training with the women subset: PPV for our model (PPV-Our) and Wang's (PPV-W) in women



(b) Training with the men subset: PPV for our model (PPV-Our) and Wang's (PPV-W) in men

Fig. 10: Comparison of the performance in HS event detection in women and men for our model and Wang's [15]. The first subfigure shows performance results from models trained on women data, and also validated on women data. The second subfigure is analogous, showing results for a different set of models trained on men data, and also validated on men data. Results are shown for each leg of each subject; for example, the label 6FL stands for the left leg of the sixth subject in Table I.

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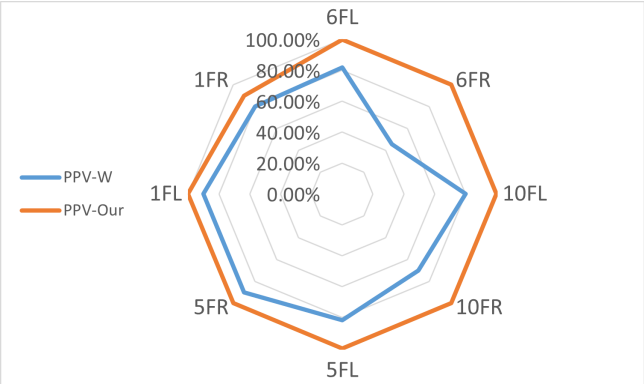
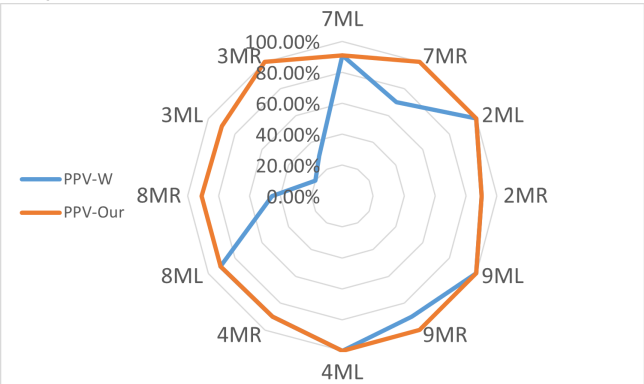


(a) Training with all data: PPV for our model (PPV-Our) and Wang's (PPV-W) in women



(b) Training with all data: PPV for our model (PPV-Our) and Wang's (PPV-W) in men

Fig. 11: Comparison of the performance in HS event detection in women and men for our model and Wang's [15]. In this figure, all data is used to train a single set of models. The first subfigure shows the results of validating these models on women data, and the second subfigure on men data. Compare with Figure 10, where each subfigure shows results from a different set of models.

the results presented an average detection rate of 78.82%, with an average detection latency of approximately 40 ms.

Further works pretend to expand this method to detect additional events and refine the kinematic gait model that specialists use. The availability of this model will enable specialists to conduct more comprehensive and specific analyses of impairments. The system is expected to evolve to identify critical points in the kinematic model related to gait impairments, adapting to the characteristics of each patient being analyzed.

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